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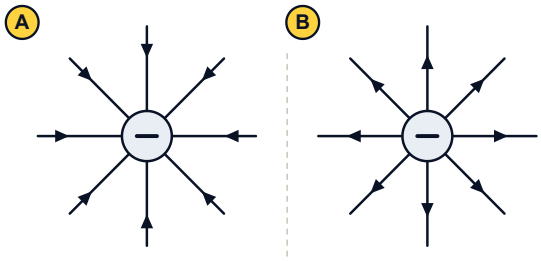
Static charge and electric fields

Electricity
Paper 1 · Separate Physics only
19 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 9 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe how insulating materials can become electrically charged.**	By rubbing them together: friction transfers electrons between them.	☐
2 ★	**Describe how a material can become negatively charged.**	By gaining electrons.	☐
3	**Describe how a material can become positively charged.**	By losing electrons.	☐
4 ★	**Describe what happens when two charged objects are brought close together.**	They exert a force on each other: like charges repel, opposite charges attract.	☐
5	**State whether electrostatic force is a contact force or non-contact force.**	Non-contact force	☐
6	**Explain why an insulator has no overall charge before it is rubbed.**	It has equal amounts of positive and negative charge.	☐
7	**Name the type of field that surrounds every charged object.**	An electric field	☐
8	**State where the electric field around a charged object is strongest.**	Closest to the object.	☐
9 ★	**Describe how the strength of an electric field changes with distance from the charge.**	The field gets stronger closer to the charge.	☐
10 ★	**State the direction in which electric field lines point.**	Away from positive charges and towards negative charges.	☐

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	QUESTION	ANSWER	✓
11	**Which diagram, A or B, shows the electric field around a negative charge? Explain.** 	A. The field lines point towards the negative charge.	☐
12	**Explain why a charged object experiences a force when close to another charged object.**	The electric field of each object exerts a force on the other object.	☐
13	**State how the size of the force between two charged objects can be increased.**	By moving them closer together.	☐
14	**State what happens when two objects with the same type of charge are brought together.**	They repel each other.	☐
15	**State what happens when two objects with opposite charges are brought together.**	They attract each other.	☐
16 ★	**A plastic rod becomes negatively charged when it is rubbed with a cloth. State and explain the charge on the cloth.**	Positive (equal in size): the cloth lost the electrons that the rod gained.	☐
17 ★	**What is an electric field?**	The region around a charged object where another charged object experiences a force.	☐
18 ★	**Describe the electric field around an isolated positively charged sphere.**	Straight lines pointing radially outwards, away from the sphere.	☐
19 ★	**Explain how a spark is produced between a charged object and an earthed conductor.**	The strong electric field ionises the air, so charge can flow across the gap.	☐

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